Credit Pulse - Targeted Personal Loan Marketing Model

Machine Learning: Personal Loan Campaign
PGP-AIML-BA-UTA-Mar25-E

06/01/2025

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Contents / Agenda

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Business Problem Overview and Solution Approach

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Analyzing customer data and building ML model provided valuable insights into customer behavior, below are the key findings and actionable recommendations for business

- Customer can be segmented in two groups earning below 90-100K, and above
- The marketing team can ignore low income group, as they don't tend to take personal loan
- To expand the asset customer base, businesses should look for high-earning, high credit card spend customers, as they are most likely to use personal loans
- High earners, large families, and highly educated customers tend to use personal loans more frequently, and the marketing team should focus on targeting these customer segments.
- Only 6% of customers use Securities Accounts, and 10% use CD Accounts, which presents an opportunity for the business to grow.
- Around 40% of customers still use physical banking, and promoting online banking presents a great opportunity for the bank to explore.

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Business Problem Overview and Solution Approach

Depending on the marketing stage, the decision tree model can be tuned as follows

- **Awareness**: Cast a wide net to build board reach and target maximum reachable customer base
- **Interest**: Narrow the focus on prospective asset customers and eliminate waste on customers unlikely to become asset customers
- **Opportunity**: This is a critical phase, and efforts should be made to ensure that no qualified leads are missed, with a focus on recall value.

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Business Problem Overview and Solution Approach

- All Life Bank wants to grow its loan book by converting “liability” customers (depositors) into “asset” customers (borrowers). Earlier marketing campaign was very successful and achieved 9% conversion rate.
- The business tasked the data science team with building an AI model that will identify potential asset customers, enabling the marketing team to target its efforts on high-potential customers
- Optimize the model, considering limited marketing resources and the cost of missing true loan prospects.

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Business Problem Overview and Solution Approach

As the business has labeled data from previous campaigns and needs to classify potential versus non-potential customers, a Decision Tree Classification ML model will be built, given its effectiveness for such use cases. We will use the following steps to build the model

- Load customer data and verify its successful loading through a sanity check
    - record size
    - number of columns and data type
    - check some sample data rows
- Perform univariate and multivariate data analysis.
- Identify and fix data anomalies found in the data.
- Build a Decision Tree Classification model that identifies customers with a high propensity to become asset customers.

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Business Problem Overview and Solution Approach

- Decision Tree is prone to overfitting and build a complex tree that captures data along with the noise so it performs very well with training data but perform poorly on test data
- Overfitting can be address by Pre pruning and Post pruning technique.
- Pre pruning is simple and faster and create simple tree structure. This is approach is much cost-effective for large data set
- Post pruning offer flexibility to fully explore the data before applying pruning but for large dataset it is costly and time-consuming

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Business Problem Overview and Solution Approach

- Develop and compare Three decision-tree–based classification models as below
  - Model with no hyperparameter tuning (default)
  - Model with pre-pruned approach (limit depth, split size and size of nodes)
  - Model with Post-pruned approach (optimizes cost-complexity and performance)
- Evaluate each model using cross-validation on Accuracy, Recall, Precision, and F1-score
- Confusion matrix and Feature importance and tree structure
- Select the configuration that best meets the business objective

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Business Problem Overview and Solution Approach

- Models are compared on Accuracy, Precision, Recall and F1 score
- In this Problem, primary focus will be on to maximize Recall and secondary focus will be on precision.
- Recall will ensure that we don’t miss any potential customer that can be converted in to asset customer and Precision will help us to reduce waste of marketing resources
- also perform checks during model tuning to guard against overfitting.

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EDA Results

- **Data Overview**
- Data had 5000 rows and 14 columns with 1 float and 13 integer columns

```
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   ID                  5000 non-null   int64  
 1   Age                 5000 non-null   int64  
 2   Experience          5000 non-null   int64  
 3   Income              5000 non-null   int64  
 4   ZIPCode             5000 non-null   int64  
 5   Family              5000 non-null   int64  
 6   CCAvg               5000 non-null   float64
 7   Education           5000 non-null   int64  
 8   Mortgage            5000 non-null   int64  
 9   Personal_Loan       5000 non-null   int64  
 10  Securities_Account  5000 non-null   int64  
 11  CD_Account          5000 non-null   int64  
 12  Online              5000 non-null   int64  
 13  CreditCard          5000 non-null   int64  
dtypes: float64(1), int64(13)
```

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EDA Results

Sample Data

	ID	Age	Experience	Income	ZIPCode	Family	CCAvg	Education	Mortgage	Personal_Loan	Securities_Account	CD_Account	Online	CreditCard
0	1	25	1	49	91107	4	1.6	1	0	0	1	0	0	0
1	2	45	19	34	90089	3	1.5	1	0	0	1	0	0	0
2	3	39	15	11	94720	1	1.0	1	0	0	0	0	0	0
3	4	35	9	100	94112	1	2.7	2	0	0	0	0	0	0
4	5	35	8	45	91330	4	1.0	2	0	0	0	0	0	1

Top 5 rows

	ID	Age	Experience	Income	ZIPCode	Family	CCAvg	Education	Mortgage	Personal_Loan	Securities_Account	CD_Account	Online	CreditCard
4995	4996	29	3	40	92697	1	1.9	3	0	0	0	0	1	0
4996	4997	30	4	15	92037	4	0.4	1	85	0	0	0	1	0
4997	4998	63	39	24	93023	2	0.3	3	0	0	0	0	0	0
4998	4999	65	40	49	90034	3	0.5	2	0	0	0	0	1	0
4999	5000	28	4	83	92612	3	0.8	1	0	0	0	0	1	1

Bottom 5 rows

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EDA Results

- Statistical summary of the data

	count	mean	std	min	25%	50%	75%	max
ID	5000.0	2500.500000	1443.520003	1.0	1250.75	2500.5	3750.25	5000.0
Age	5000.0	45.338400	11.463166	23.0	35.00	45.0	55.00	67.0
Experience	5000.0	20.104600	11.467954	-3.0	10.00	20.0	30.00	43.0
Income	5000.0	73.774200	46.033729	8.0	39.00	64.0	98.00	224.0
ZIPCode	5000.0	93169.257000	1759.455086	90005.0	91911.00	93437.0	94608.00	96651.0
Family	5000.0	2.396400	1.147663	1.0	1.00	2.0	3.00	4.0
CCAvg	5000.0	1.937938	1.747659	0.0	0.70	1.5	2.50	10.0
Education	5000.0	1.881000	0.839869	1.0	1.00	2.0	3.00	3.0
Mortgage	5000.0	56.498800	101.713802	0.0	0.00	0.0	101.00	635.0
Personal_Loan	5000.0	0.096000	0.294621	0.0	0.00	0.0	0.00	1.0
Securities_Account	5000.0	0.104400	0.305809	0.0	0.00	0.0	0.00	1.0
CD_Account	5000.0	0.060400	0.238250	0.0	0.00	0.0	0.00	1.0
Online	5000.0	0.596800	0.490589	0.0	0.00	1.0	1.00	1.0
CreditCard	5000.0	0.294000	0.455637	0.0	0.00	0.0	1.00	1.0

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EDA Results

Statistical summary -

- IDs span 1–5,000 and will be dropped as it has unique value for each row.
- Age: average 45 years min age from 23 to max age of 67
- Experience: mean ~20 years, but a few negative values (min –3) suggest data entry issues;
- Mortgage: over half have zero mortgage (median 0), but some high outliers (max 635).
- Personal_Loan 9.6% have taken loan, which is consistent with previous marketing initiative
- Securities_Account and CD_Account ownership is very small and could be a business opportunity to increase customer base for these products
- Only 60% customer use the online banking

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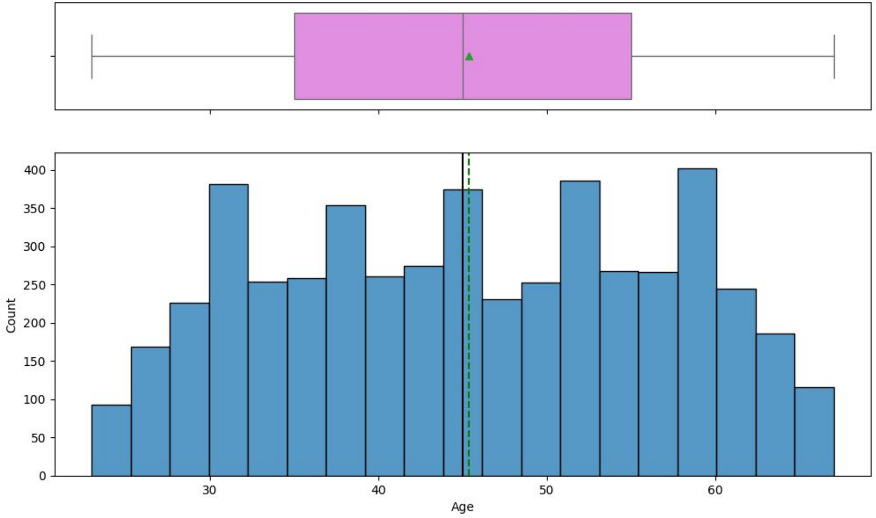
EDA Results

- **Age** has median around 45 years and range from 23 to 67 and well spread across the range. 75% of the customers are below 55 years of age



Age Histogram with Box plot

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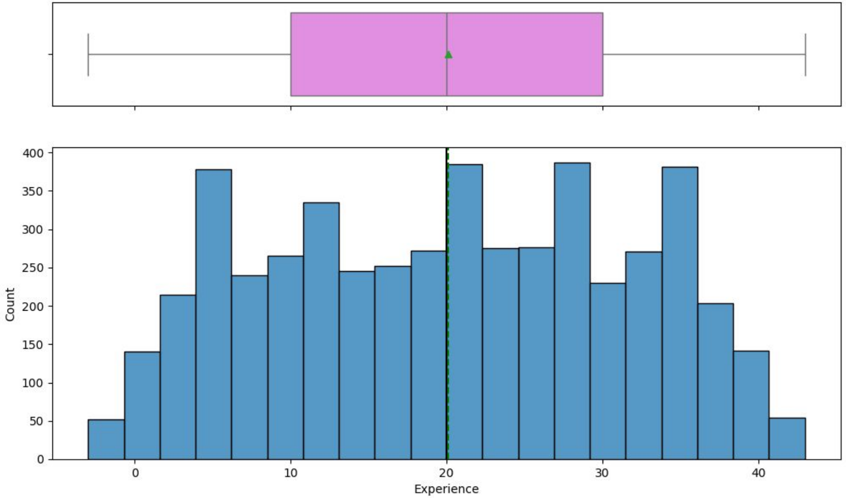
EDA Results

- **Experience** has median around 20 years and range from 0 to 43
- Data is well distributed across the range of experience.



Experience box plot with histogram

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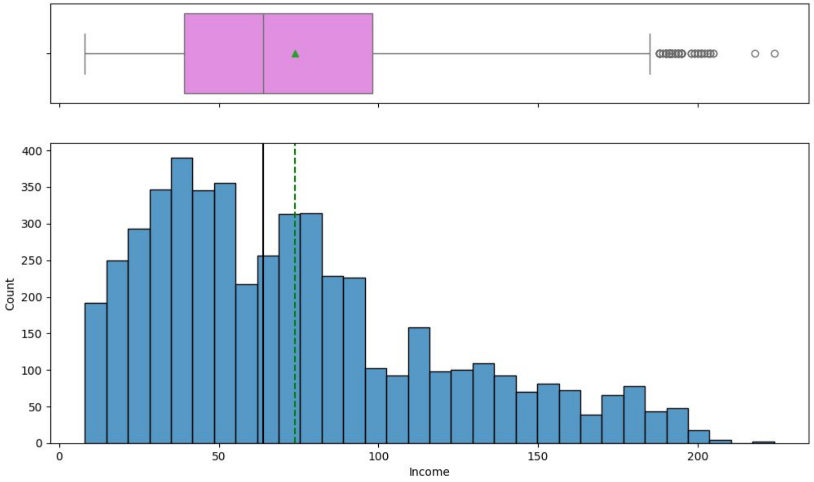
EDA Results

- **Income** data is right skewed, with mean 73K and range from 8K to 224. Most of the customers income is below 98K. It indicates bank cater from very low earners to high earners data has many outliers too



Income box plot with box plot

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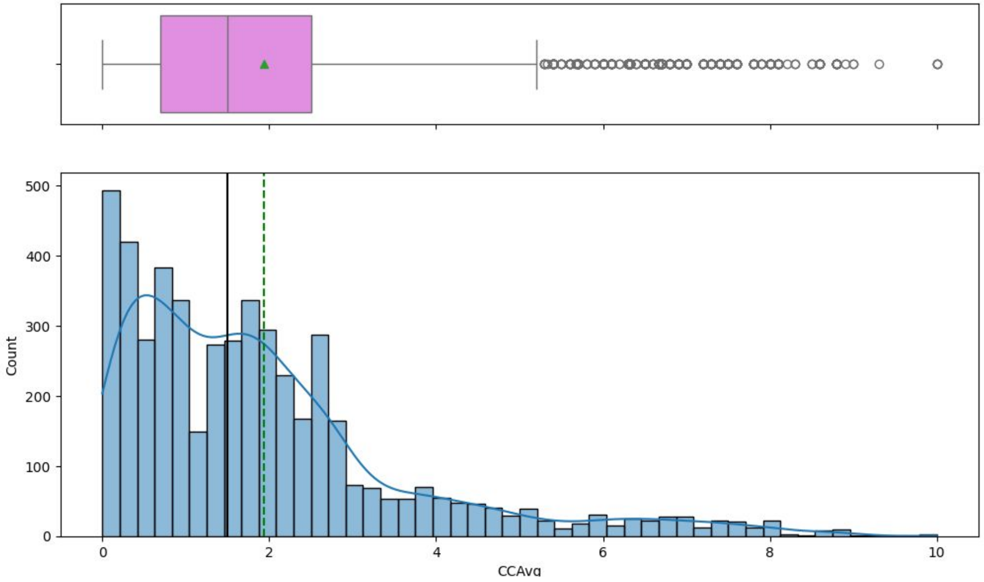
EDA Results

- **CCAvg**: Credit card average spend is around 2K with 75 % spend is below 2.5K there are few outliers for high spenders that go up to 10k.



Credit card average monthly spend box blot with histogram

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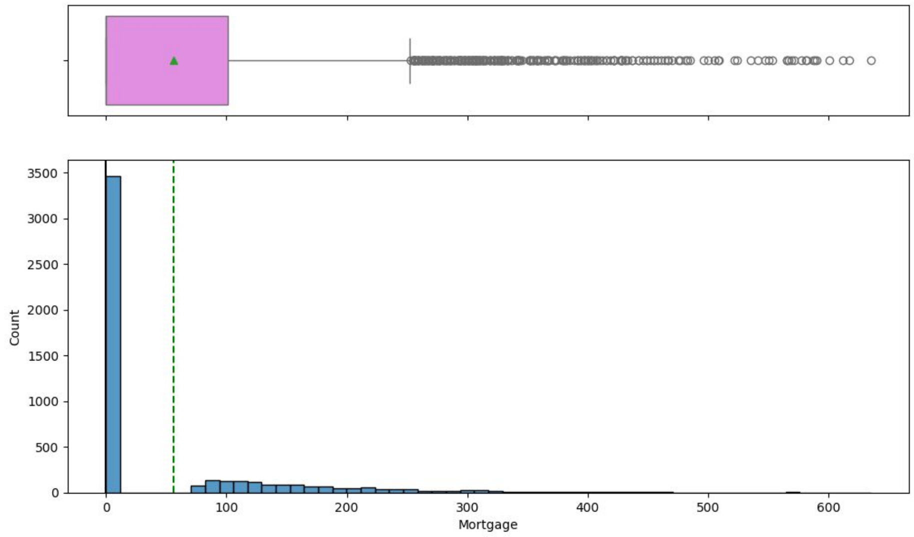
EDA Results

- **Mortgage -** 69% of the customers don't have the mortgage or data is missing, so mean is impacted by it to be around 0 and most of the values below 101K



Mortgage box plot with histogram (with 0 values)

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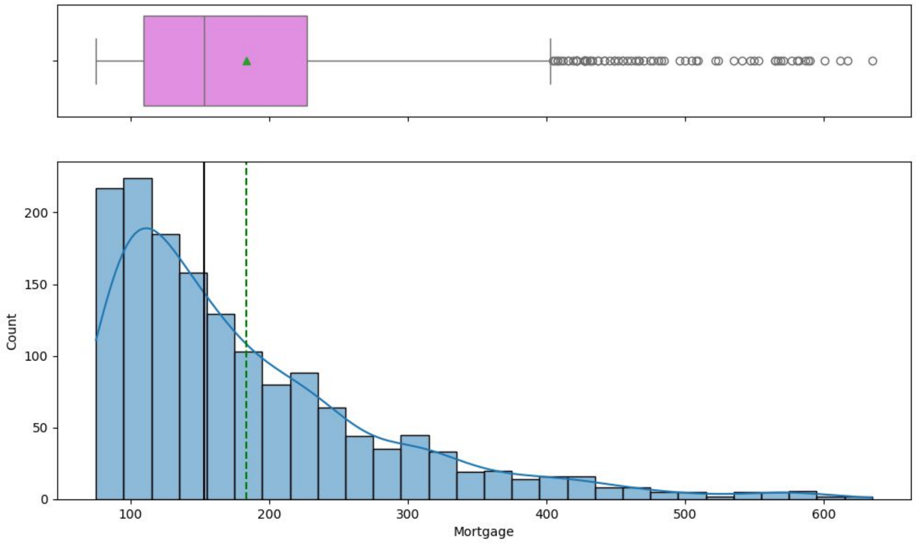
EDA Results

- When filtering the data to customers with mortgages, the mean loan amount is around $153K, with 75% of loans below $227K. However, there are large outliers, with the maximum loan amount reaching up to $635K
- Data is right skewed to indicate most of the mortgage values are below 230K



Mortgage box plot with histogram (with filtering out 0 values)

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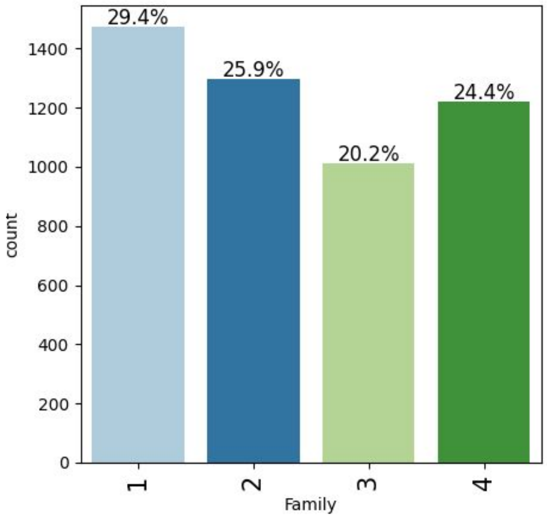
EDA Results

- **Family** size ranges from 1 to 4 and is well distributed across this range
- Single-person households comprise 29.4% (1,472) of the customer base
- Couples and families of four have comparable representation, with 25.9%(1,296) and 24.4% (1,222) customers, respectively
- Family size of 3 is least represented at 20.2% (1010)



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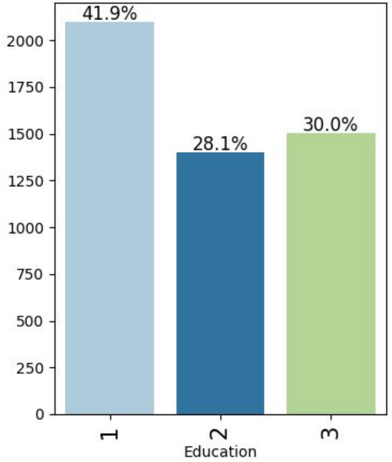
EDA Results

- **Education** is well-represented across undergraduate, graduate, and advanced/professional categories, with most customers holding an undergraduate degree. The graduate category is the least represented group compared to the others.



Education Bar plot

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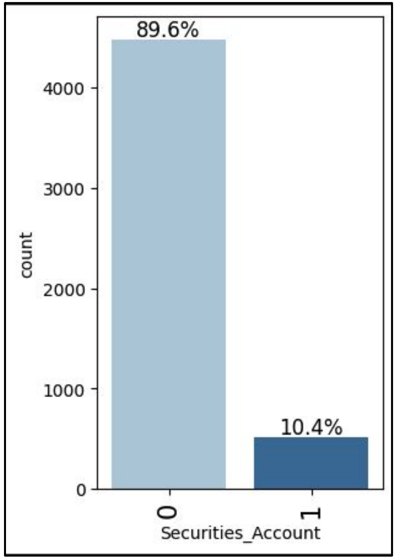
EDA Results

- **Security account** is held by a small proportion of customers, with only about 10% (approximately 500 customers) having security accounts. The business may consider expanding its customer base by promoting security account products.



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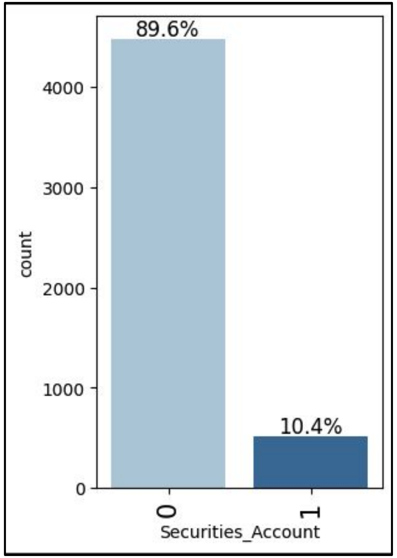
EDA Results

- **CD accounts** are held by a small proportion of customers, with around 6% (approximately 300 customers) having CD accounts. Like security accounts, CD account products also have significant growth potential



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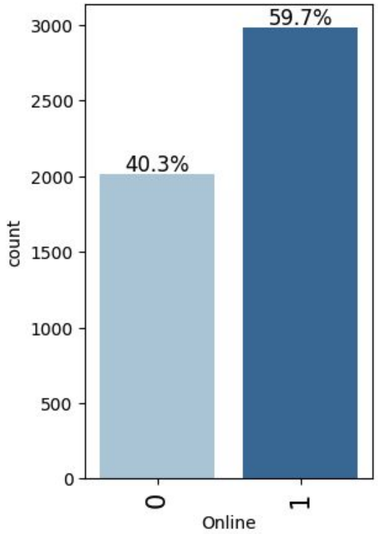
EDA Results

Online banking is popular among many customers, a significant portion still relies on physical banking services. Specifically, around 60% of customers use online banking, while approximately 40% prefer physical banking. To optimize service delivery, efforts should be made to increase online banking usage and address any hurdles preventing customers from adopting digital banking



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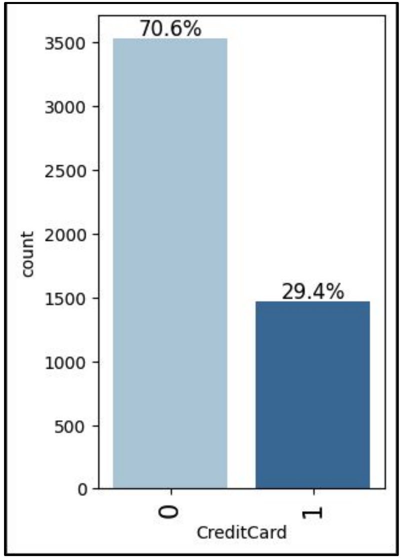
EDA Results

- **Credit Card** - Approximately 70% of customers don't have credit cards, while around 30% of customers use them. It would be worthwhile to explore opportunities for the bank to expand its credit card offerings through targeted marketing



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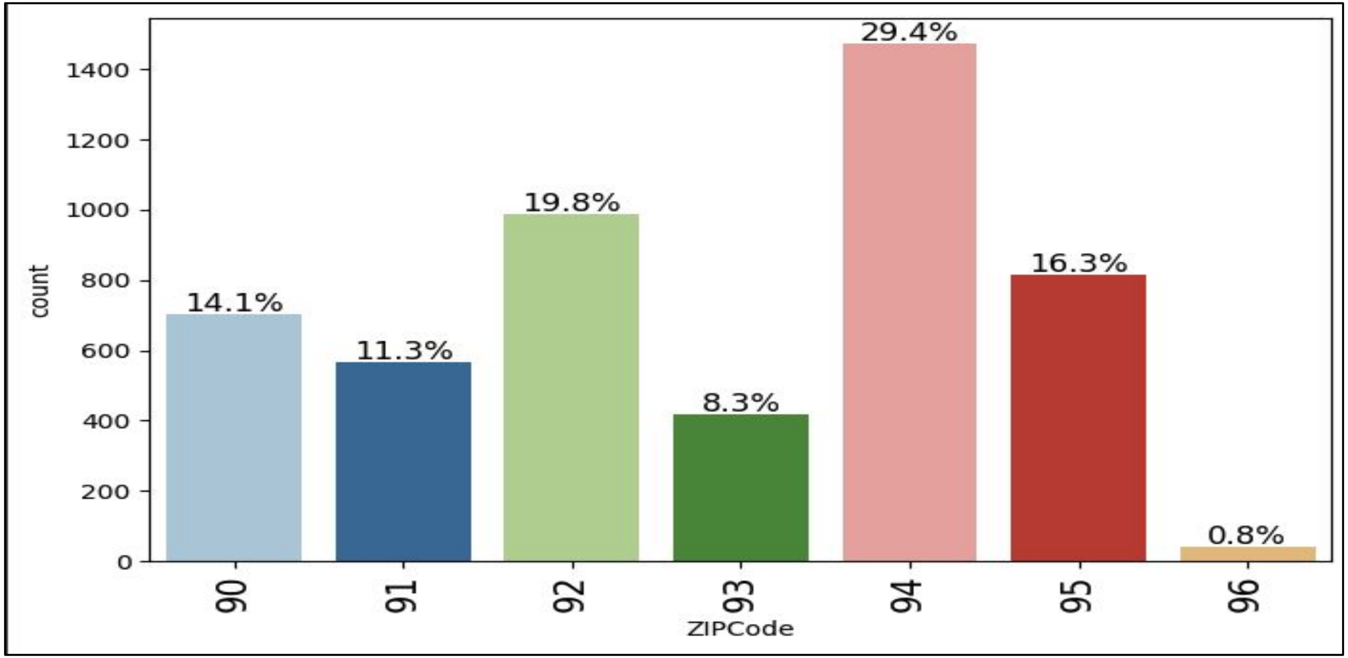
EDA Results

- Customers are concentrated in few of the Zip code. Zip code starting with 94 account for 30% customer and zip code 92XX around 20% and zip code 95XX and 90 XX are around 15 %. Zip code 96XX has the least amount of customer and bank can concentrate effort to increase its costumer base in 93XX, 96XX zip codes



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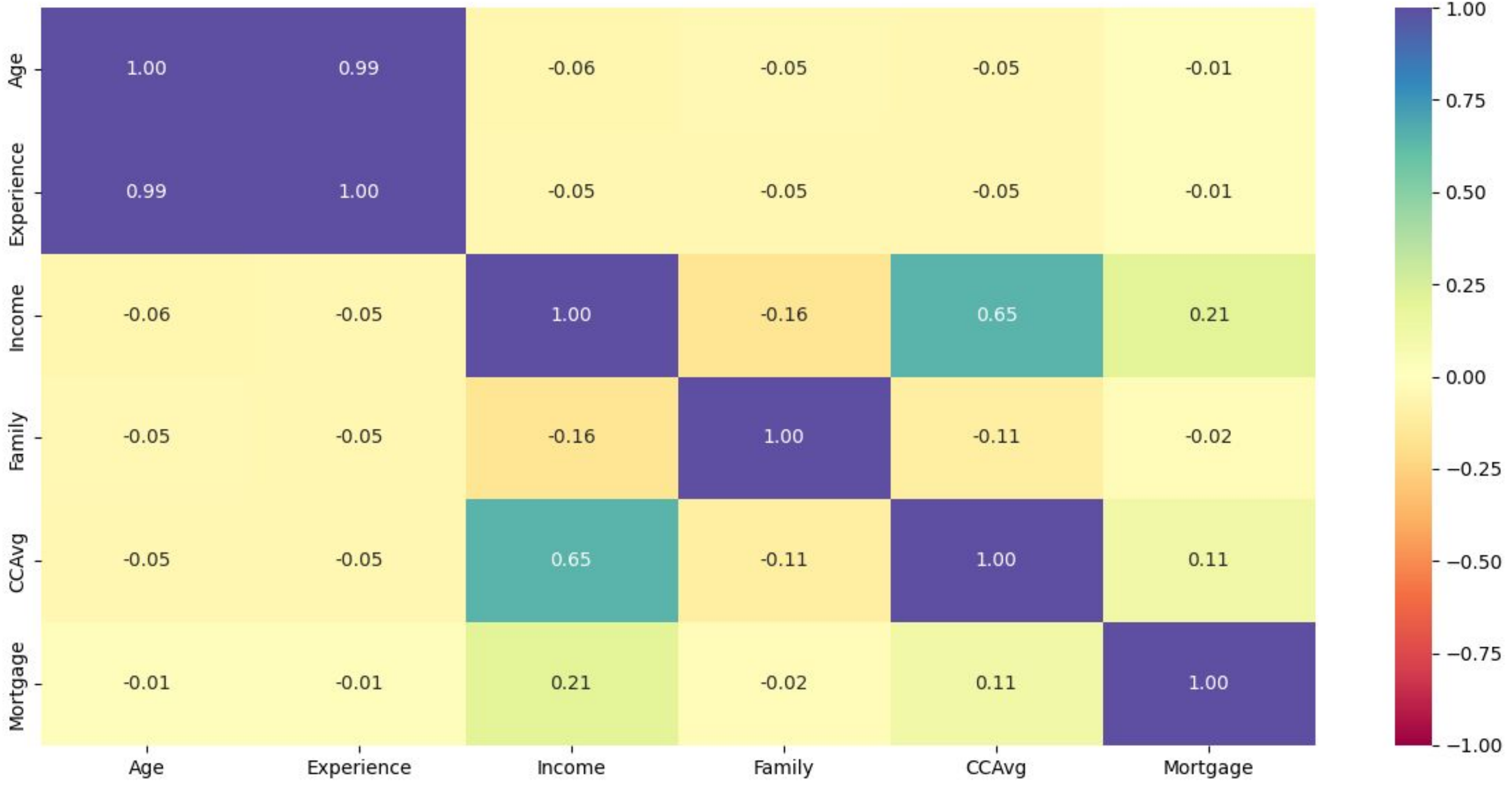
EDA Results

Correlation heat Map



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EDA Results

- **Bivariate Analysis**
- Heat map show correlation between various features,
- Education and Age are directly correlated and convey same data, so we can drop Age are representative of Education
- Income and Credit card averages has high correlation.
- Customer with Higher income tend to spend more or have higher monthly average spend
- Other features have very weak correlation and represent independent features

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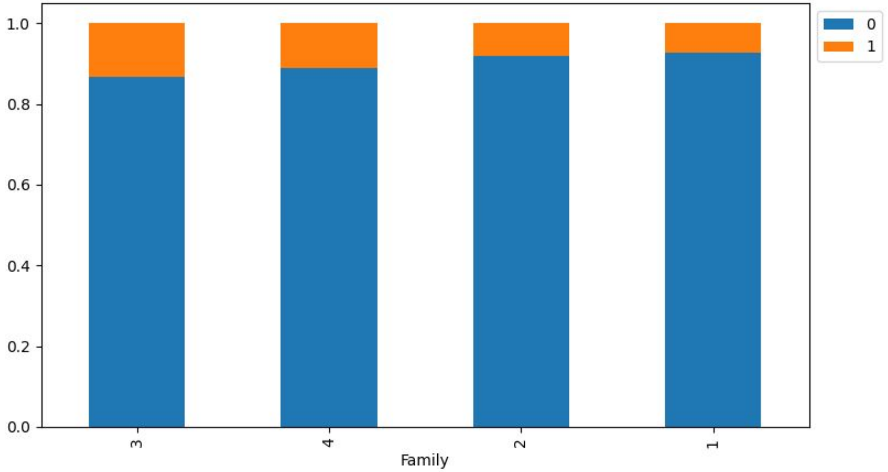
EDA Results

- Large family usually used personal loan more compare the smaller family size
- 13% of family size 3 customer used personal loan
- 10% of the customer with family size of 4 used personal loan
- 7-8% of the customer with family size of 1 or 2 used personal loan



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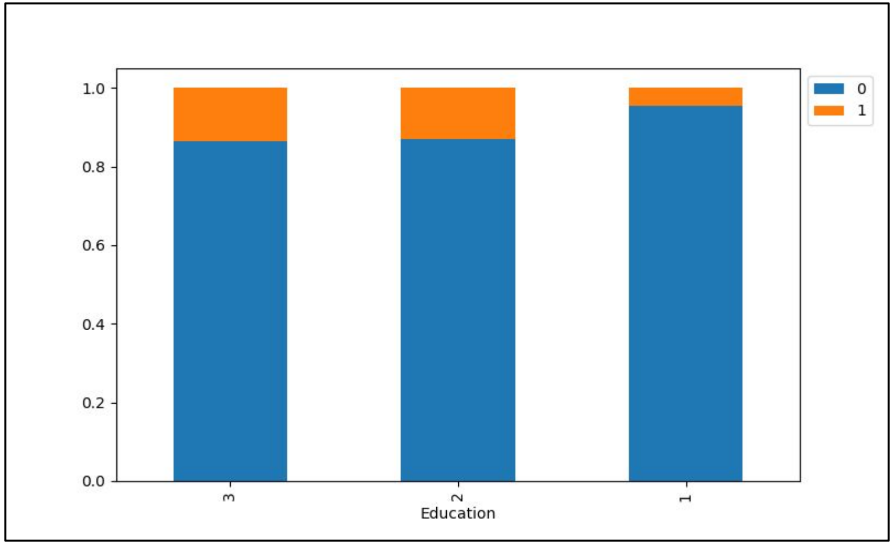
EDA Results

- Customers with graduate and advanced/professional education are more likely to take out personal loans. In contrast, undergraduate customers tend not to opt for personal loans
- The data shows that around 13% of advanced/professional customers and 12% of graduate customers have taken personal loans, compared to only 4% of undergraduate customers."



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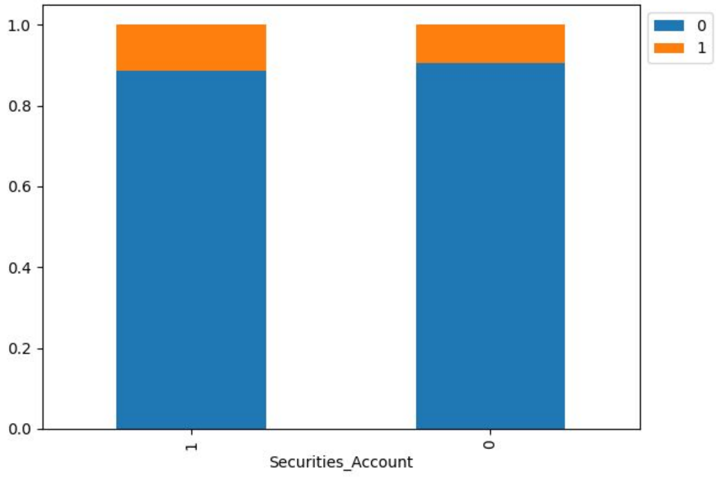
EDA Results

- Personal account conversation is quite similar for customer with security account or customer with no security account. Data suggest security account don't much have correlation to asset customer
- 11% customer with security account converted to asset customer
- 9% customer with no security account were converted to asset customers.



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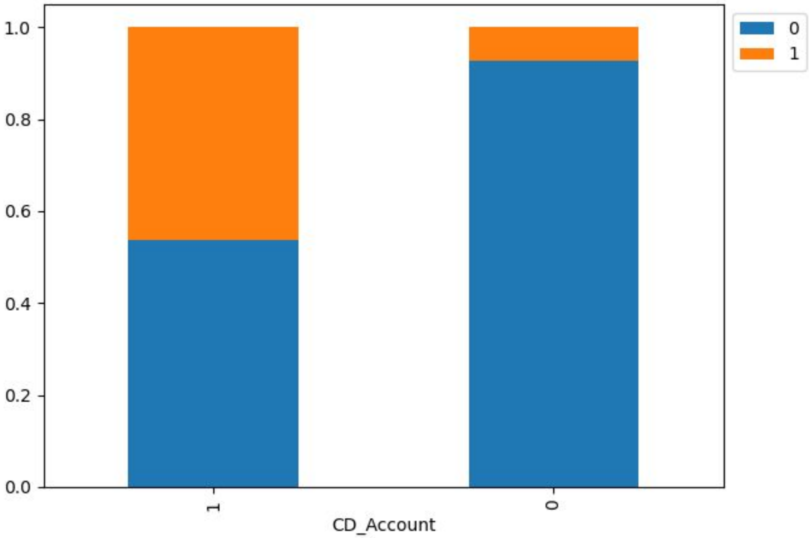
EDA Results

- Around half of the customer with CD account used Personal loan vs only 7% customer used personal loans with no CD account.
- We can infer that customer with CD account most likely use the personal loan



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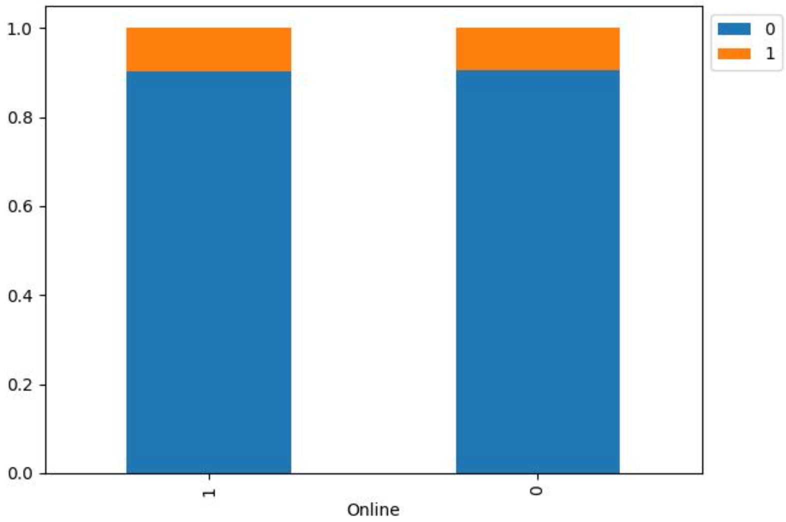
EDA Results

- Online and physical banking customer don't show much of difference in availing the personal loan, around 9% of both the customer type used the Personal banking. So we can ignore Online feature from considering for the model building



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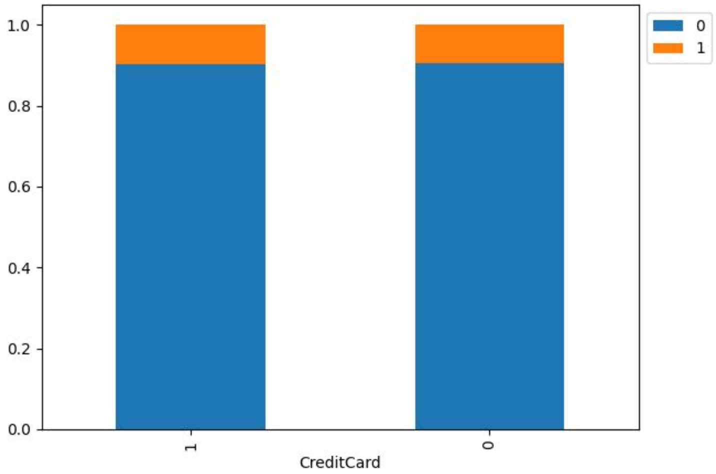
EDA Results

- Customer with credit cards or no credit cards don’t show much of difference in availing the personal loan, there is 9% of customer in each category that use personal loan



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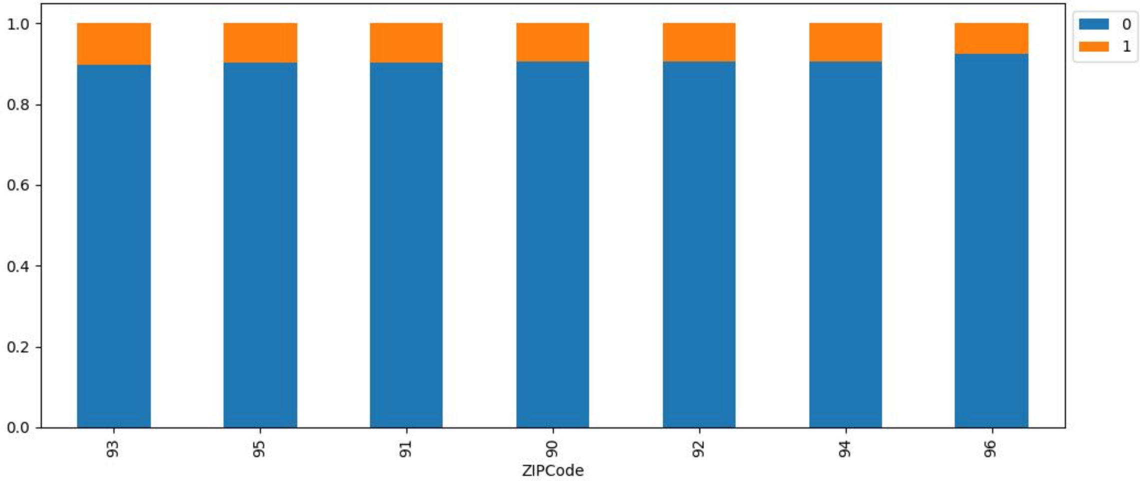
EDA Results

- Zip code did not show much distinction in conversation of customer to asset group most of the zip code show similar conversion rate around 9/10 % except zip code 96XX around 7%.



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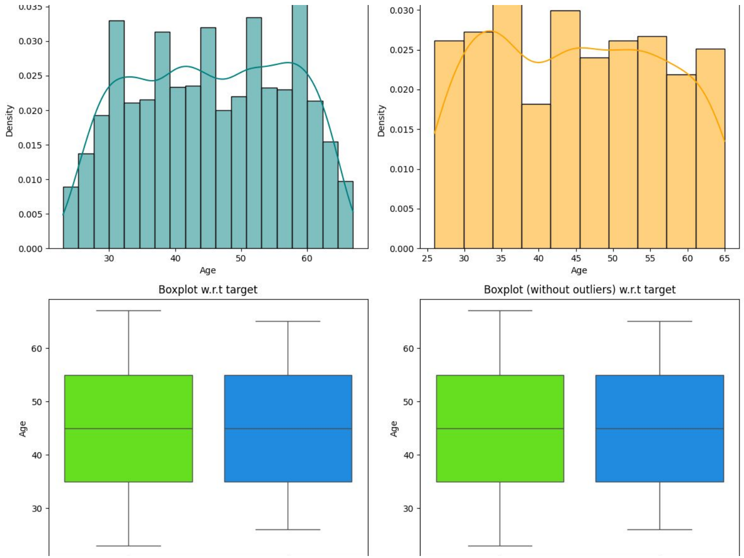
EDA Results

- Age and Personal loan have very weak relations ship, and we can see that age has very low indication on Customer conversion to asset customer.
- Asset and non asset customer mean age and 25, 50, 75 Percentile data matches.



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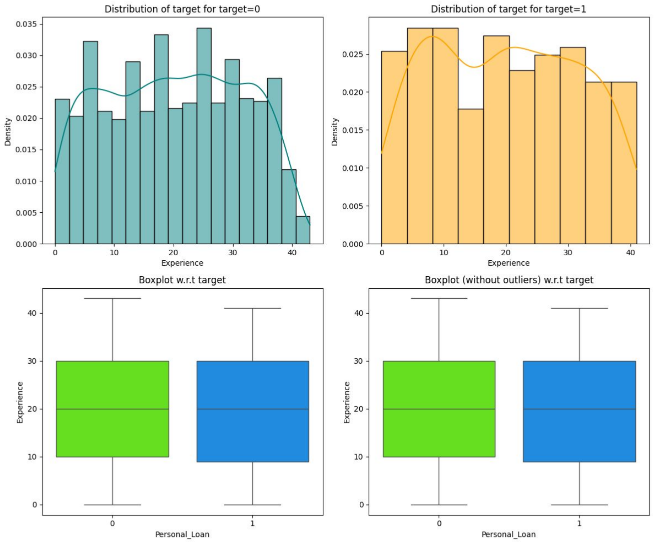
EDA Results

- Experience and Personal loan have very weak relations ship, and we can see that age has very low indication on Customer conversion to asset customer. With both median around 20 years experience



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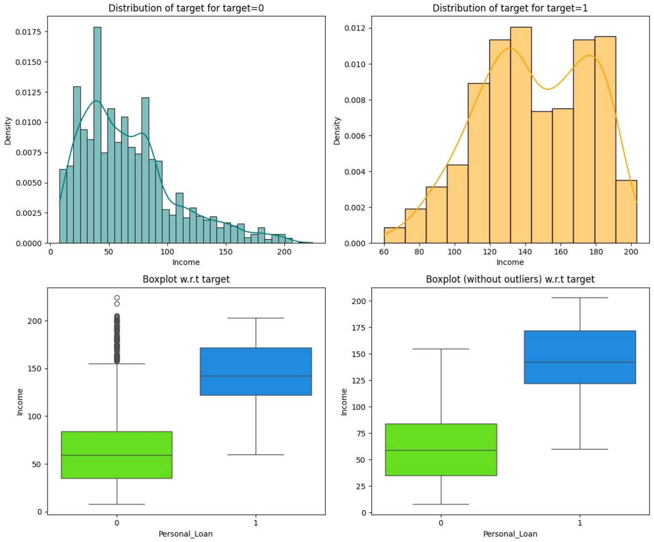
- Higher income customers used personal loan. 140K is the mean income of the customer using Personal loan with and without outlier having personal loans. Customer with income below 90k mostly don’t used personal loan.



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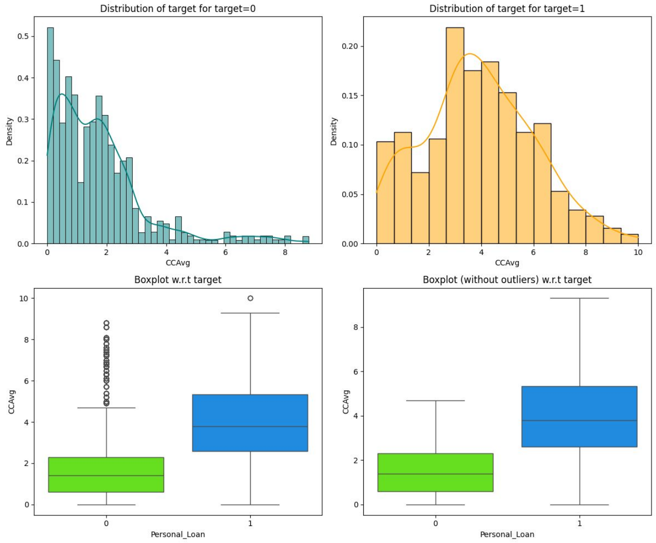
- **CCAvg -**Customers with a monthly average credit card spend below $2,200 tend not to have personal loans, while those with an average spend above $2,500 are more likely to use personal loans. This suggests that customers with higher credit card spending habits are more likely to utilize personal loans and its very important feature



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Data Preprocessing

- No duplicate or missing values found
- Below are the % outlier found for various columns under considerations which are very small amount of outlier and seems valid data point, so no action needed.

	0
Age	0.00
Experience	0.00
Income	1.92
Family	0.00
CCAvg	6.48
Mortgage	5.82

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Data Preprocessing

- ID column contains only unique values and dropped from the dataset
- Experience columns had few negative values to that are fixed by replacing those values to positive values
- Zip code data had high cardinality, so it was trimmed to the first two digits, resulting in 7 unique values that represent larger geographic zones.
- Following columns type converted to categorical columns Education, Personal_loan, Securities account, CD Account, Online, Credit Card, Zip Code.
- Dropped the Experience columns as it is perfectly correlated with Age
- One hot encoding applied to Zip code and Education columns to prepare data for model building

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Data Preprocessing

- Data was split into train and test sets with 70:30 ratio. Random State =1 seed value used to achieve repeatable result across multiple runs. Verified that Personal loan has same weightage

Data	Personal Loan = 0	Personal Loan = 1
Train	0.905429	0.0945721
Test	0.900667	0.099333

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Model Building

- Model is built using DecisionTreeClassifier with criteria = gini and random state = 1
- Gini impurity quantifies the probability of incorrectly classifying a randomly chosen element from the node, based on the class distribution within that node
- Model Accuracy, recall and F1 score is compared for Test and Train data
- Compared Test and train data performance on various combination of tree depth, max leaf nodes and min sample size
- Compared Test and train data on various alpha values
- Visualize Confusion matrix created for Test and Train model results
- Built function to visualize Decision Tree, and analyze how model computed the decision
- Build Function to compare the performance of all three models

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Model Building

Great Learning POWER AHEAD

- Default Model with no pruning or no hyperparameters tuning applied
- Pre-Prune Model with hyperparameter, tuned based on the best recall improvements
- Post-Prune Model with cost-complexity Pruning parameter, consider recall values
- Visualize and analyze decision tree build by the model
- Verified features used by model and their weightage in evaluation criteria

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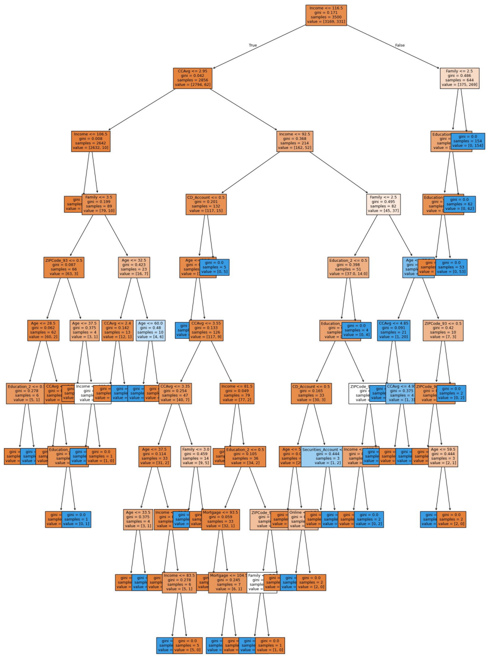
Model Performance Summary

- **Default Model Tree** is very complex and tried to pick up all the data and noise points. It's very difficult to explain the model outcome from the tree structure

```
|--- Income <= 116.50
|   |--- CCAvg <= 2.95
|   |   |--- Income <= 106.50
|   |   |   |--- weights: [2553.00, 0.00] class: 0
|   |   |--- Income >  106.50
|   |   |   |--- Family <= 3.50
|   |   |   |   |--- ZIPCode_93 <= 0.50
|   |   |   |   |   |--- Age <= 28.50
|   |   |   |   |   |   |--- Education_2 <= 0.50
|   |   |   |   |   |   |   |--- weights: [5.00, 0.00] class: 0
|   |   |   |   |   |   |--- Education_2 >  0.50
|   |   |   |   |   |   |   |--- weights: [0.00, 1.00] class: 1
|   |   |   |   |   |--- Age >  28.50
|   |   |   |   |   |   |--- CCAvg <= 2.20
|   |   |   |   |   |   |   |--- weights: [48.00, 0.00] class: 0
|   |   |   |   |   |   |--- CCAvg >  2.20
|   |   |   |   |   |   |   |--- Education_3 <= 0.50
|   |   |   |   |   |   |   |   |--- weights: [7.00, 0.00] class: 0
|   |   |   |   |   |   |   |--- Education_3 >  0.50
|   |   |   |   |   |   |   |   |--- weights: [0.00, 1.00] class: 1
|   |   |   |   |--- ZIPCode_93 >  0.50
|   |   |   |   |   |--- Age <= 37.50
|   |   |   |   |   |   |--- weights: [2.00, 0.00] class: 0
|   |   |   |   |   |--- Age >  37.50
|   |   |   |   |   |   |--- Income <= 112.00
|   |   |   |   |   |   |   |--- weights: [0.00, 1.00] class: 1
|   |   |   |   |   |   |--- Income >  112.00
|   |   |   |   |   |   |   |--- weights: [1.00, 0.00] class: 0
|   |   |   |--- Family >  3.50
|   |   |   |   |--- Age <= 32.50
|   |   |   |   |   |--- CCAvg <= 2.40
|   |   |   |   |   |   |--- weights: [12.00, 0.00] class: 0
|   |   |   |   |   |--- CCAvg >  2.40
|   |   |   |   |   |   |--- weights: [0.00, 1.00] class: 1
|   |   |   |   |--- Age >  32.50
|   |   |   |   |   |--- Age <= 60.00
|   |   |   |   |   |   |--- weights: [0.00, 6.00] class: 1
|   |   |   |   |   |--- Age >  60.00
|   |   |   |   |   |   |--- weights: [4.00, 0.00] class: 0
|   |--- CCAvg >  2.95
|   |   |--- Income <= 92.50
|   |   |   |--- CD_Account <= 0.50
|   |   |   |   |--- Age <= 26.50
|   |   |   |   |   |--- weights: [0.00, 1.00] class: 1
|   |   |   |   |--- Age >  26.50
```



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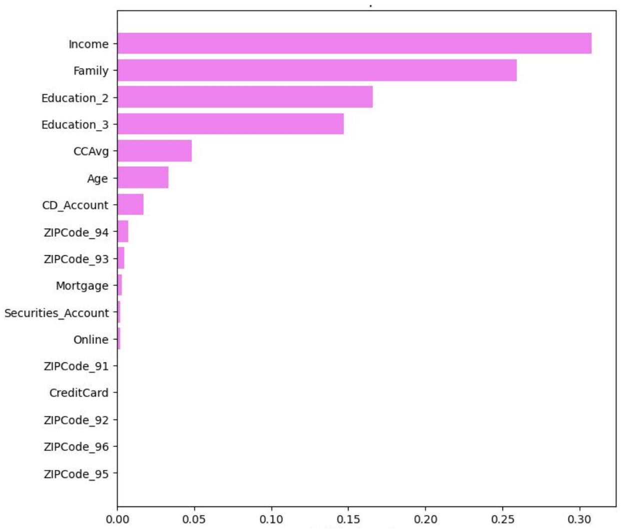
Model Performance Summary

- **Default** Model used many features in decision criteria, and it is very difficult to explain or get insight using them and suggest very complex decision model



	Imp
Income	0.308098
Family	0.259255
Education_2	0.166192
Education_3	0.147127
CCAvg	0.048798
Age	0.033150
CD_Account	0.017273
ZIPCode_94	0.007183
ZIPCode_93	0.004682
Mortgage	0.003236
Securities_Account	0.002224
Online	0.002224
ZIPCode_91	0.000556
CreditCard	0.000000
ZIPCode_92	0.000000
ZIPCode_96	0.000000
ZIPCode_95	0.000000

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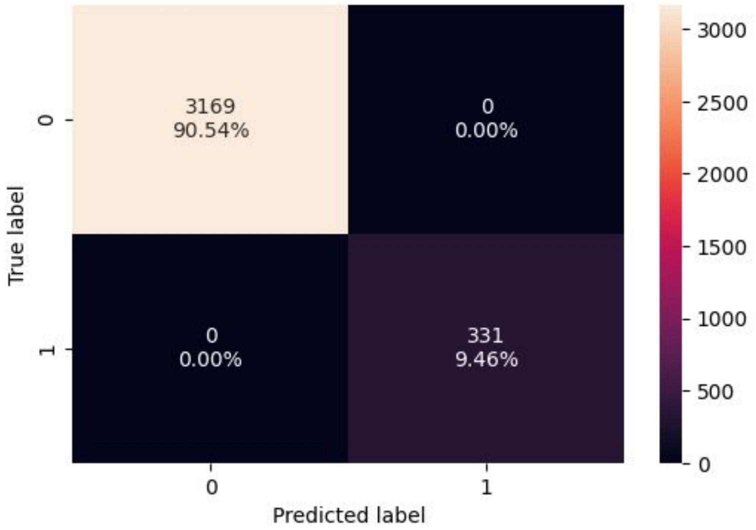
Model Performance Summary

- **Default Model** -Confusion Matrix on Train data shows model had 0 false positive and 0 false negative This indicates on training data model has perfect Accuracy, Precision, recall and F1 score



Accuracy	Recall	Precision	F1
1.0	1.0	1.0	1.0

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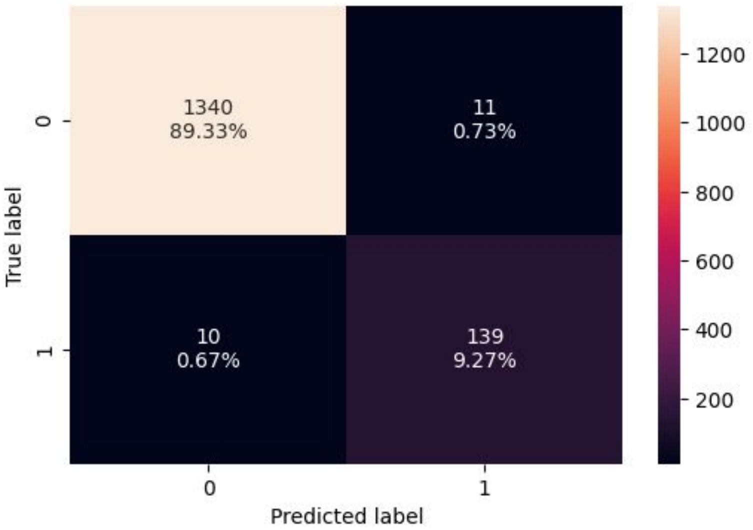
Model Performance Summary

- **Default Model** - Confusion Matrix on Test data reveals 1(0.73%) False positives and 10 (0.67%) False negative, indicating slightly inferior performance on the test data. This suggests that the Model is overfitted and need further investigation and tuning to mitigate the issue



Accuracy	Recall	Precision	F1
0.986	0.932886	0.926667	0.929766

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Model Performance Summary

- **Pre-Pruning Model:** various values of Max depth, max leaf node and max split size used to determine optimal pre-pruning depth
- To determine optimal value, minimal difference in recall score between train and test data is used
- Below Hyper parameters are used to create the Pre-Prune Model
  DecisionTreeClassifier(class_weight='balanced', max_depth=np.int64(2), max_leaf_nodes=50, min_samples_split=1, random_state=42)

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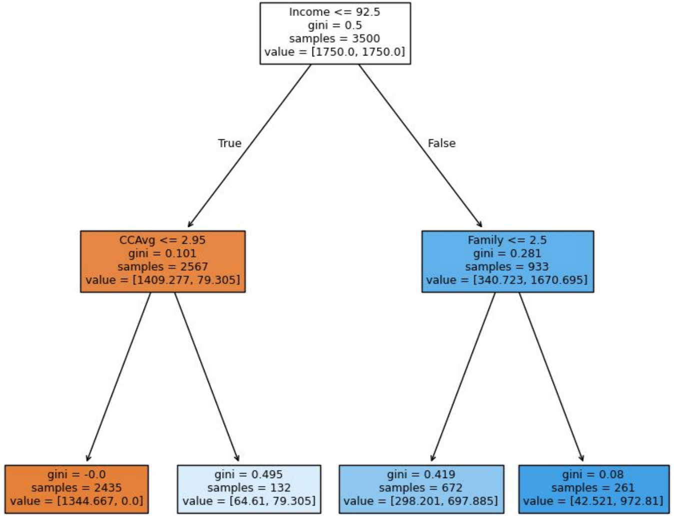
Model Performance Summary

- **Pre-Prune Model** Tree is very simple and only consider Income, CCAvg and Family size.

```
|--- Income <= 92.50
|   |--- CCAvg <= 2.95
|   |   |--- weights: [1344.67, 0.00] class: 0
|   |--- CCAvg >  2.95
|   |   |--- weights: [64.61, 79.31] class: 1
|--- Income >  92.50
|   |--- Family <= 2.50
|   |   |--- weights: [298.20, 697.89] class: 1
|   |--- Family >  2.50
|   |   |--- weights: [42.52, 972.81] class: 1
```



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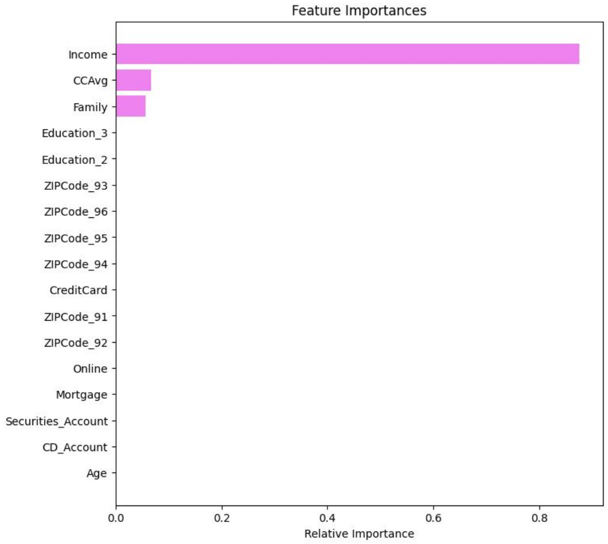
Model Performance Summary

- **Pre-Pruning Model:** Model Decision is very easy to explain with Income, credit card average spend and Family size.



	Imp
Income	0.876529
CCAvg	0.066940
Family	0.056531
Age	0.000000
Mortgage	0.000000
Securities_Account	0.000000
CD_Account	0.000000
Online	0.000000
CreditCard	0.000000
ZIPCode_91	0.000000
ZIPCode_92	0.000000
ZIPCode_93	0.000000
ZIPCode_94	0.000000
ZIPCode_95	0.000000
ZIPCode_96	0.000000
Education_2	0.000000
Education_3	0.000000

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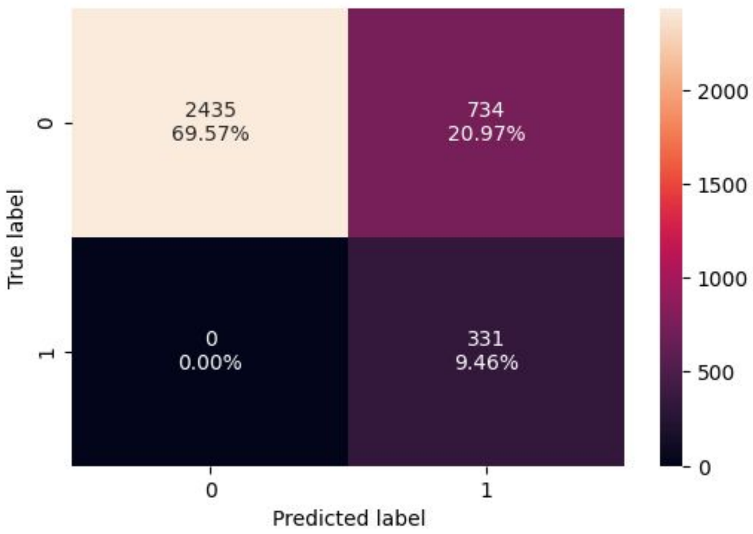
Model Performance Summary

- **Pre-Pruning Model:** Confusion matrix and performance on the training data show perfect recall with 0 false negatives. However, the model predicted numerous false positives, which negatively impacted accuracy, precision, and F1 score



Accuracy	Recall	Precision	F1
0.790286	1.0	0.310798	0.474212

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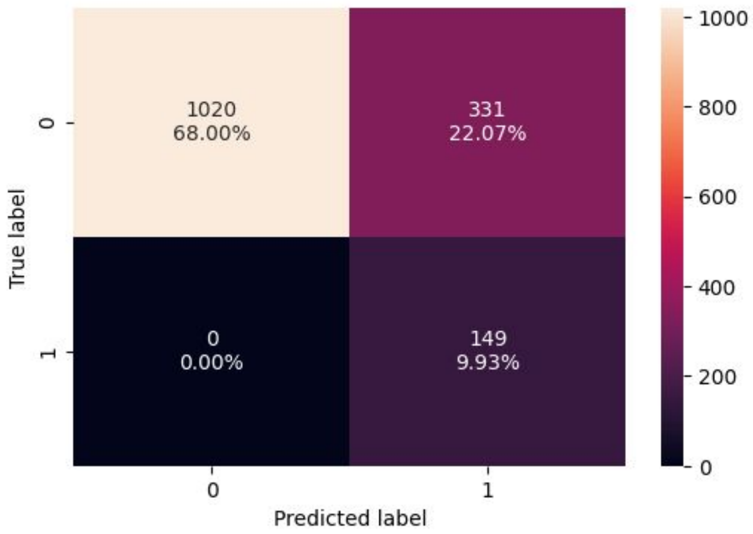
Model Performance Summary

- Confusion matrix and model performance on the Test Data shows perfect recall with 0 false negatives, but Model predicted large False Positive values. Notably, the accuracy decreased compared to the training data, indicating potential overfitting or generalization issues



Accuracy	Recall	Precision	F1
0.779333	1.0	0.310417	0.473768

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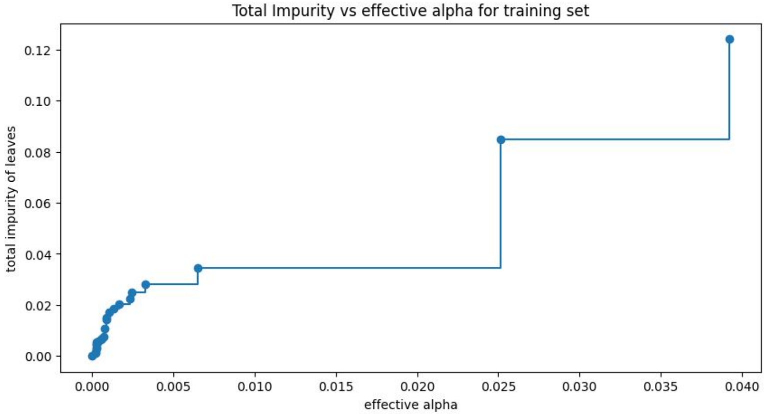
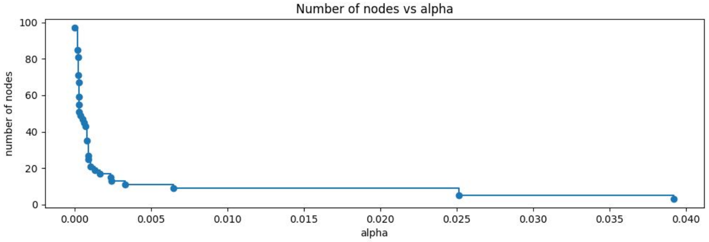
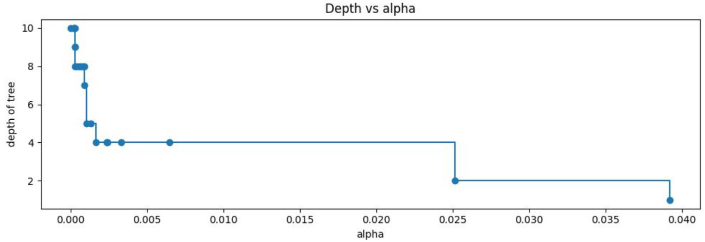
Model Performance Summary

- **Post-Pruning Model**: evaluated based on the Alpha and impurities values. When alpha is 0 tree overfits and 0 impurity leaves and Number of nodes/ Depth becomes very large







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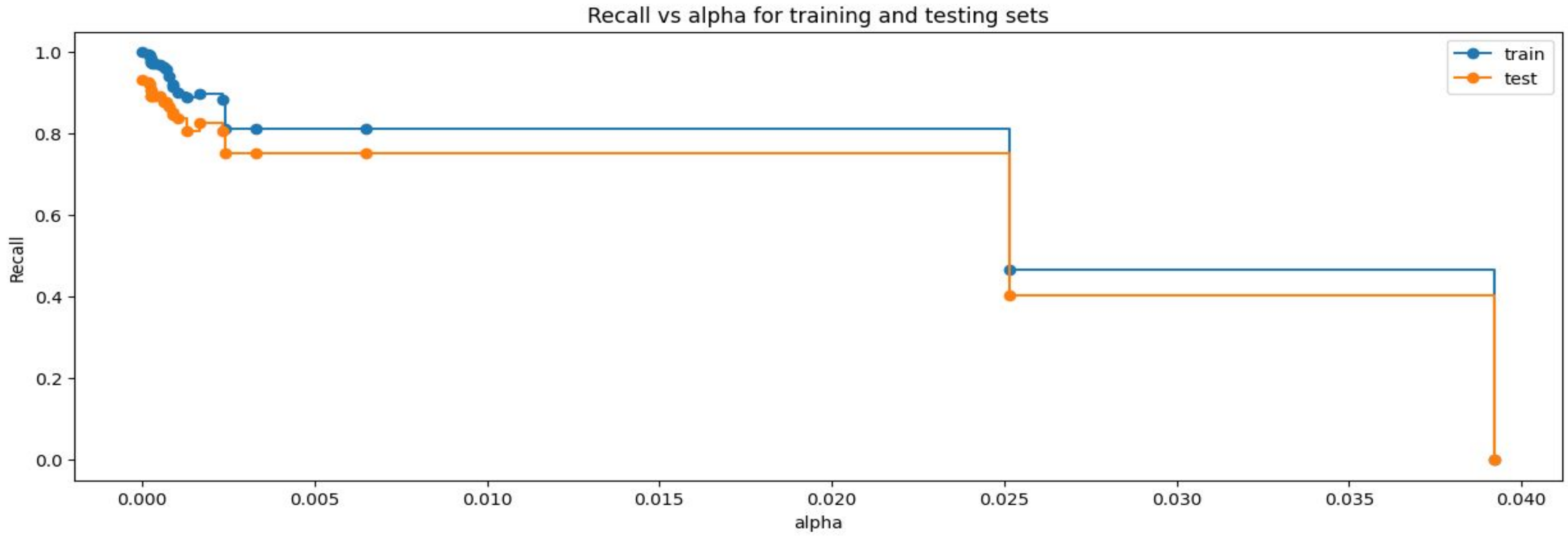
Model Performance Summary

- **Post-Pruning Model**: Compared recall with Alpha value for both train and test data to chek for optimal alpha values



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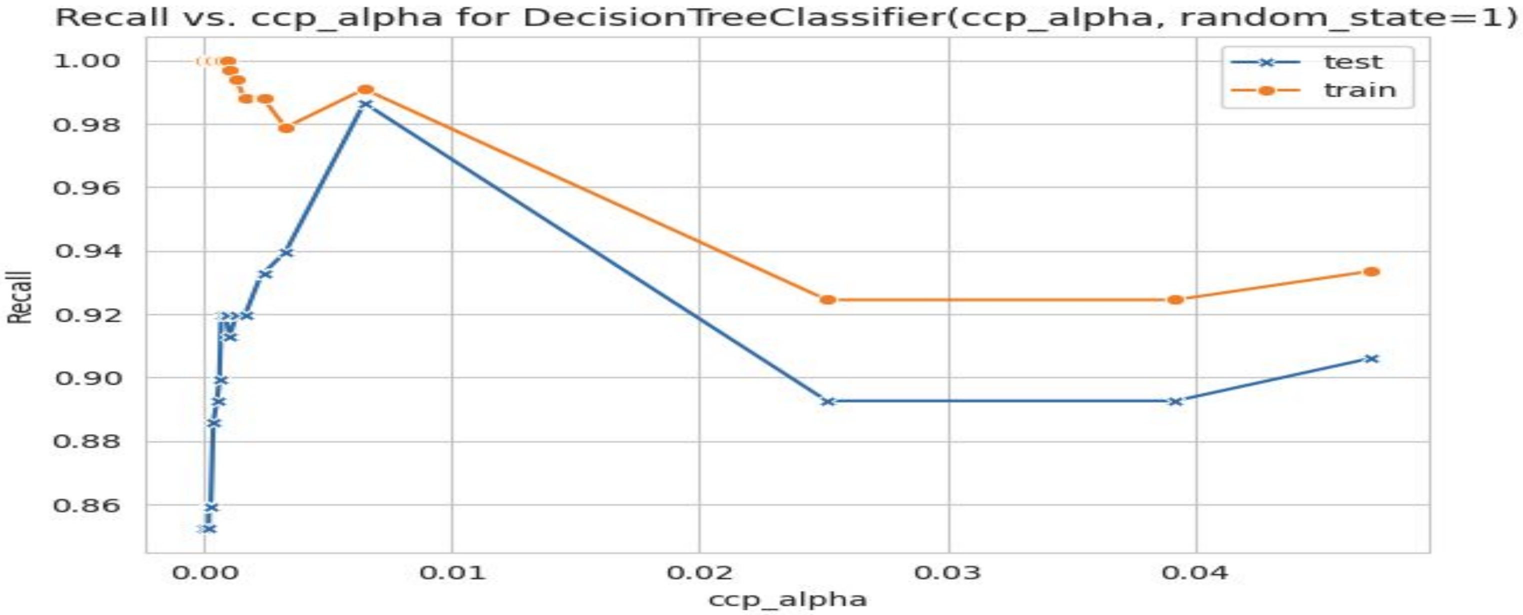
Model Performance Summary

- **Post-Pruning Model**: optimal alpha found 0.**0064728147182238****1** **where** train and test recall scored gap is minimal.
  DecisionTreeClassifier(ccp_alpha=0.**0064728147182238****11**, class_weight={0: 0.15, 1: 0.85}, random_state=1)



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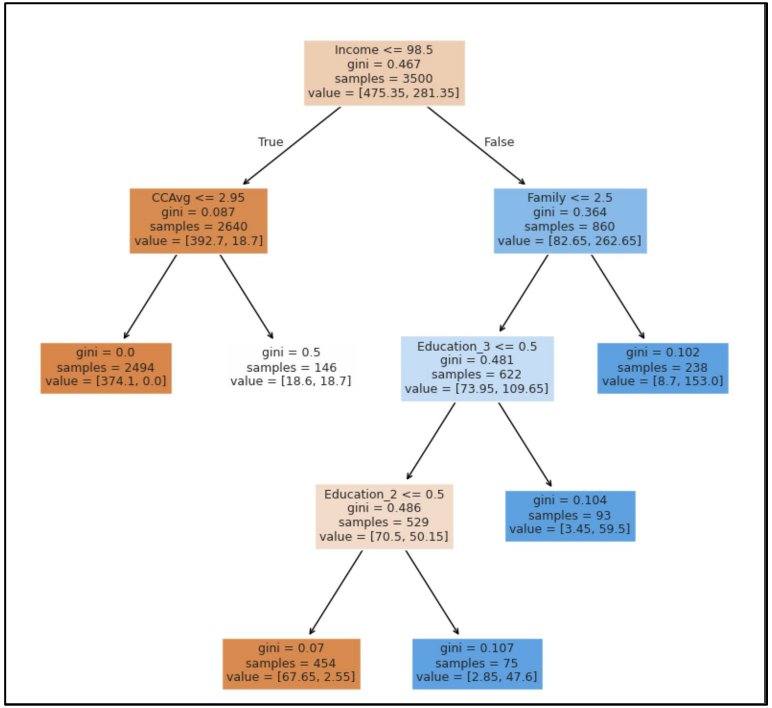
Post Prune

- **Post-Prune Model**: Tree is slightly more complex compare to pre-prune model but much simpler compare to default model also it considers very important factor in decision

```
|--- Income <= 98.50
|   |--- CCAvg <= 2.95
|   |   |--- weights: [374.10, 0.00] class: 0
|   |--- CCAvg >  2.95
|   |   |--- weights: [18.60, 18.70] class: 1
|--- Income >  98.50
|   |--- Family <= 2.50
|   |   |--- Education_3 <= 0.50
|   |   |   |--- Education_2 <= 0.50
|   |   |   |   |--- weights: [67.65, 2.55] class: 0
|   |   |   |--- Education_2 >  0.50
|   |   |   |   |--- weights: [2.85, 47.60] class: 1
|   |   |--- Education_3 >  0.50
|   |   |   |--- weights: [3.45, 59.50] class: 1
|   |--- Family >  2.50
|   |   |--- weights: [8.70, 153.00] class: 1
```



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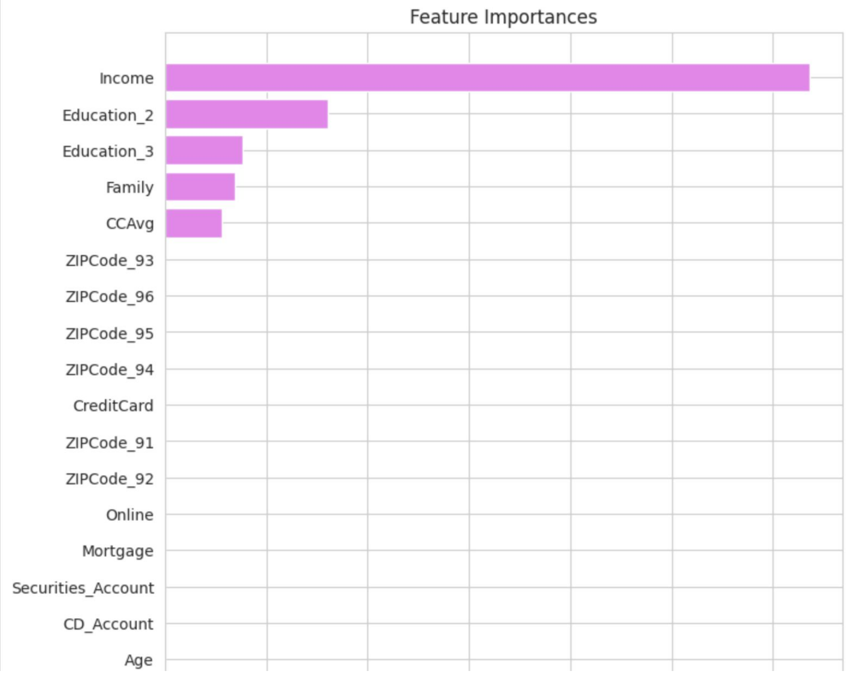
Model Performance Summary

- Post Prune Model considered Income, Education Family size and Credit card average spend. Which is much more feature compare to pre pruning model but very less compare to default model.



	Imp
Income	0.636860
Education_2	0.160224
Education_3	0.076930
Family	0.069445
CCAvg	0.056541
Age	0.000000
CD_Account	0.000000
Securities_Account	0.000000
Mortgage	0.000000
CreditCard	0.000000
Online	0.000000
ZIPCode_91	0.000000
ZIPCode_92	0.000000
ZIPCode_94	0.000000
ZIPCode_93	0.000000
ZIPCode_96	0.000000
ZIPCode_95	0.000000

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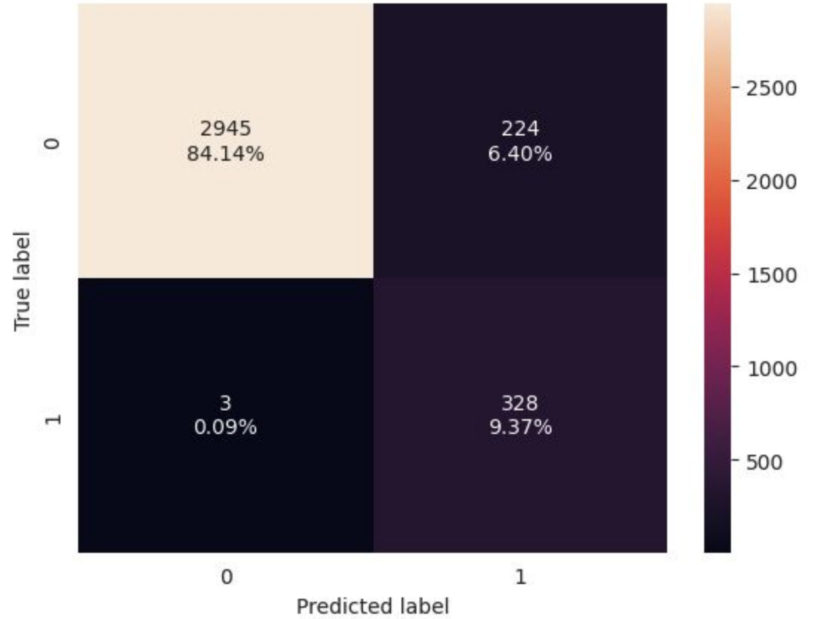
Model Performance Summary

- **Post-Pruning Model:** Confusion matrix and Model performance on the Train Data indicate 2.86% false negative and 0.71% False positive, and it had better accuracy, Precision and F1 score compare to Pre-prune model



	Accuracy	Recall	Precision	F1
0	0.935143	0.990937	0.594203	0.742922

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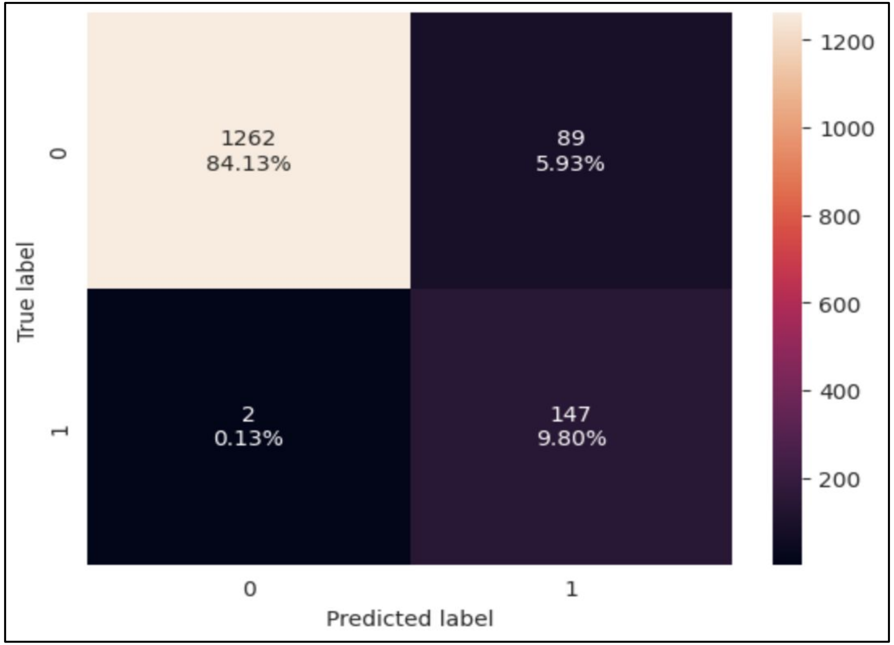
Model Performance Summary

- **Post-Pruning Model:** Confusion matrix and model performance on the Test Data indicate there were 2.60% False Negative and 1.07% false positive. Model recall was Lower than the train data. but better precision and F1 score than training set



	Accuracy	Recall	Precision	F1
0	0.939333	0.986577	0.622881	0.763636

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Model Performance Improvement

- Compare All the models on their training and test data performance

Training performance comparison:

	Decision Tree (sklearn default)	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	1.0	0.790286	0.935143
Recall	1.0	1.000000	0.990937
Precision	1.0	0.310798	0.594203
F1	1.0	0.474212	0.742922

Test set performance comparison:

	Decision Tree (sklearn default)	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.986000	0.790286	0.939333
Recall	0.932886	1.000000	0.986577
Precision	0.926667	0.310798	0.622881
F1	0.929766	0.474212	0.763636

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Model Performance Improvement

- **Default model**
  - This model performed very well on train data
  - It considered a large set of feature
  - Decision Tree was very complex, and difficult to interpret the decision
  - noise along with the data intricacies
  - This model performance deteriorated on test data indicates overfitting and model captured noise along with the data.
- **Pre Prune Model**
  - Model performance similar on training and test data, indicating model is tuned well
  - Model achieved best Recall score =1 for training and train means this will not miss any potential customer
  - Model precision score = 0.319921 is very low and indicates it has many false positive and will waste some marketing resources

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Model Performance Improvement

- **Post Prune Model**
  - Model performance similar on training and test data, indicating model is not overfitting
  - Model achieved very good Recall score =0.986577 for training and train means this will not miss any potential customer
  - Model precision score = 0.622881 is quite improved compare to pre-pruning
  - The tree structure of post pruning is slightly complex but over all model fits fair well on most of the aspect.

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Model Performance Improvement

- Overall, the post-pruning model performed well on most evaluation criteria, optimizing both recall and precision. For businesses aiming to optimize budget with minimal loss of opportunity, the post-pruning model is best suited
- If the primary focus is to target all potential asset customers without missing any, even if it means spending more on non-potential asset customers, the pre-pruning model is a better fit
- Pre-pruning model can be used in the initial awareness phase, and later switched to the post-pruning model for more optimized spending, depending on business needs

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APPENDIX

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